30 March 1976
SUPERSEDING
(see 6.5).

MILITARY SPECIFICATION

CLOTH, LAMINATED, AND CLOTH, COATED (CHLOROPRENE COATED)

This specification is approved for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers the requirements for four types and two classes of laminated and coated cloths, to be used in the fabrication of double-wall air-supported shelter tents (see 6.1).

1.2 Classification. The laminated and coated cloths shall be of the following types and classes as specified (see 6.2).

Type I - Air retaining wall cloth
  Class 1 - Olive Green 207 - Exterior Wall
  Class 2 - White - Interior Wall

Type II - Double texture floor cloth

Type III - Web cloth - air-retaining

Type IV - Web cloth - non air-retaining

2. APPLICABLE DOCUMENTS

2.1 Issues of documents. The following documents of the issue in effect on date of invitation for bids or request for proposal, form a part of this specification to the extent specified herein.

SPECIFICATIONS

FEDERAL

L-P-1040 - Plastic Sheets and Strips (Polyvinyl Fluoride)
PPP-P-1136 - Packaging and Packing of Coated (Plastic; Rubber) and Laminated Fabrics

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MIL-C-43286 - Cloth, Plain Weave, Nylon; Cloth, Plain Weave, Polyester

STANDARDS

FEDERAL

FED-STD-191 - Textile Test Methods
FED-STD-406 - Plastics; Methods of Testing
FED-STD-595 - Colors
FED-STD-601 - Rubber; Sampling and Testing

MILITARY

MIL-STD-105 - Sampling Procedures and Tables for Inspection by Attributes
MIL-STD-282 - Filter Units, Protective Clothing, Gas-Mask Components and Related Products: Performance - Test Methods
MIL-STD-1487 - Glossary of Cloth Coating Imperfections

(Copies of specifications, standards, drawings and publications required by contractors in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

3. REQUIREMENTS

3.1 Laboratory report approval. Unless otherwise specified at the time of submission of a bid (see 6.2), the bidder shall submit to the contracting officer a certified copy of a recent laboratory report covering material which he proposes to deliver. Unless otherwise specified by the contracting officer, the bidder shall certify that the material was manufactured in a plant where the coating will be performed if a contract is awarded. The laboratory report shall contain test data which demonstrates that the finished product which the contractor proposes to deliver has been tested and found to comply with all requirements of this specification. Any of the following types of reports will be satisfactory from the standpoint of this requirement.

a. An independent or commercial laboratory report.
b. The prospective supplier's own laboratory report.
c. A Governmental laboratory report from a contract within 6 months of date of submission of bid.

The purpose of the above requirement is to assist the Government to determine the capability of bidders to manufacture a cloth meeting all the requirements of this specification. The submission of an acceptable laboratory report under this requirement shall not be construed as relieving a contractor from subsequently meeting all requirements of this specification on all deliveries.

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3.2 Samples.

3.2.1 Standard sample. When a standard sample is available, the laminated cloth shall match the standard sample for shade and shall be equal to or better than the standard sample with respect to all characteristics for which the standard sample is referenced (see 6.3).

3.3 Material.

3.3.1 Base cloth. The base cloth shall be polyester, conforming to type I of MIL-C-43286.

3.3.2 Coating compound. The coating shall be chloroprene rubber suitably compounded and plasticized only with phthalate or phosphate ester plasticizers and pigmented to a suitable shade of black. An organic isocyanate may be added to this compound in order to achieve the adhesion of coating values specified in table I.

3.3.3 Laminating film. The film used in the laminated cloths shall be polyvinyl fluoride, conforming to type III, class 1 of L-P-1040. The grade A or B shall be as specified for each type of laminated cloth.

3.4 Application of coating compound and laminating film.

3.4.1 Type I, laminated cloths. The base cloth specified in 3.3.1 shall be coated with the black chloroprene compound specified in 3.3.2 in the following sequence:

a. Coat one side of the base cloth with 1 ± 0.25 ounces per square yard of the black compound specified in 3.3.2. This shall be the back side of the coated cloth used in the laminate.

b. Coat the other side of the cloth with 3.5 ± 0.5 ounces per square yard of the black compound specified in 3.3.2 for type I, class 1. For type I, class 2, coat the other side of the cloth with 4.5 ± 0.5 ounces per square yard of the black compound specified in 3.3.2. The heavily coated side of each class shall be the face side of the coated cloth used in the laminate.

c. The polyvinyl fluoride film specified in 3.3.3 shall then be laminated to the heavily coated face side of the coated cloth in such a manner that the finished laminate will meet the requirements in table I. The polyvinyl fluoride film for the type I, class 1 laminated cloth shall be grade A, Olive Green 207, and for the type I, class 2 laminated cloth shall be grade B, White.

3.4.2 Type II, double texture floor cloth. The base cloth specified in 3.3.1 shall be coated on each side with 1.25 ± 0.25 ounce per square yard of the black chloroprene compound specified in 3.3.2, and shall be subsequently laminated to each side of a grade A, Olive Green 207, or a grade B, white polyvinyl fluoride film specified in 3.3.3.

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3.4.3 Type III, web cloth, air-retaining. The base cloth specified in 3.3.1 shall be coated on one side with 1 ± 0.25 ounces per square yard, and on the opposite side with 3.5 ± 0.5 ounces per square yard, of the black chloroprene compound specified in 3.3.2. The finished coated cloth shall meet the requirements specified in table I.

3.4.4 Type IV, web cloth, non air-retaining. The base cloth specified in 3.3.1 shall be coated on one side only with 1 ± 0.25 ounces per square yard of the black chloroprene specified in table I.

3.5 Dusting. The use of talc or dusting compound is prohibited during manufacture, finishing, or packing of the laminated or coated cloth, for shipping. If necessary to prevent blocking during shipment, the finished laminated or coated cloths may be interleaved with polyethylene film.

3.6 Finished cloths. Both the coated and laminated cloths shall be fully vulcanized and shall conform to all the requirements specified in table I.

TABLE I. Laminated cloth and coated cloth requirements

Characteristic	Requirements				
	Type I		Type II	Type III	Type IV
	Class 1	Class 2			
Overall weight, oz./sq. yd					
Minimum	10.0	11.0	14.5	7.75	4.75
Maximum	12.0	13.0	19.0	9.75	5.75
Breaking strength, pounds (minimum)					
Warp	200	200	300	200	200
Filling	200	200	300	200	200
Tearing strength, grams (minimum)					
Initial:					
Warp	3500	3500	6000	3500	3500
Filling	3000	3000	6000	3000	3000
After accelerated weathering: 1/					
Warp	3500	3500	6000	---	---
Filling	3000	3000	6000	---	---

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TABLE I. Laminated cloth and coated cloth requirements (cont'd)

Characteristic	Requirement				
	Type I		Type II	Type III	Type IV
	Class 1	Class 2			
After oven aging:					
Warp	3500	3500	6000	3500	3500
Filling	3000	3000	6000	3000	3000
Hydrostatic resistance p.s.i. (minimum): 3/					
Initial	400	400	400	400	---
After strength of coating	300	300	300	300	---
Film adhesion, lbs./2-inch strip (minimum)					
Dry	10.0	10.0	---	---	---
Wet	10.0	10.0	---	---	---
At 180°F	4.0	4.0	---	---	---
Coating adhesion (back side) (minimum) lbs./2-inch strip					
Dry	12.0	12.0	5.0	8.0	8.0
Wet	12.0	12.0	5.0	8.0	8.0
At 180°F	4.0	4.0	3.0	4.0	4.0
Ply adhesion (type II only) (minimum)					
Dry	---	---	4.0	---	---
Wet	---	---	4.0	---	---
At 180°F	---	---	2.0	---	---
Adherability	4/	4/	4/	4/	---
Blocking at 200°F (both sides)					
Scale rating (maximum)	No. 2	No. 2	No. 2	No. 2	No. 2
Cold crack at -40°					
Hydrostatic	300	300	300	300	---
Visual	2/	2/	2/	2/	---
Air leakage test:					
Standard conditions	3/	3/	3/	3/	---
180°F	3/	3/	3/	3/	---

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TABLE I. Laminated cloth and coated cloth requirements (cont'd)

Characteristic	Requirement				
	Type I		Type II	Type III	Type IV
	Class 1	Class 2			
Resistance to chemical warfare agents, minutes (minimum)					
Initial:					
Mustard H	100	100	100	---	---
GB	200	200	200	---	---
After accelerated weathering:					
Mustard H	100	100	100	---	---
GB	200	200	200	---	---
After decontamination:					
Mustard H	100	100	100	---	---
GB	200	200	200	---	---
After cold crack:					
Mustard H	100	100	100	---	---
GB	200	200	200	---	---

1/ The laminated cloth shall show no evidence of becoming stiff, brittle, soft, tacky and there shall be no cracking or crazing when flat or when folded sharply on itself by hand, face out. In addition, there shall be no appreciable* change of the color or exudation of the plasticizer in the exposed test specimen when compared with an unexposed test specimen. (*Appreciable means a change that is noticeable at first glance when comparing tested specimen with original.)

2/ There shall be no evidence of cracking or separation of the polyvinyl fluoride film or the chloroprene coating when examined along the fold line.

3/ There shall be no evidence of blistering or delamination of the polyvinyl fluoride film or chloroprene coating.

4/ Not more than 1/16 inch slippage.

3.7 Width. The minimum overall width, without selvages, shall be as specified by the procuring activity (see 6.2). All selvages shall be trimmed from the type I and II laminated cloths, and may be trimmed from the type III and IV coated cloths.

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3.8 Length and put-up. The cloth length (all types) shall be put-up in rolls not less than 60 yards nor more than 74 yards as specified in PPP-P-1136. An individual roll shall contain not more than three pieces and no piece shall be less than 6 yards in length.

3.9 Color. The color shall match the standard sample, when available (see 3.2.1 and 6.3). When no standard sample is available, the color shall match the applicable color number of FED-STD-595.

3.9.1 Matching. The color of the finished coated cloth shall match the standard sample (see 6.3) under artificial daylight having a color temperature of 7500° Kelvin, and shall be a good approximation to the standard sample under incandescent lamplight at 2800° Kelvin.

3.10 Workmanship. The finished laminated or coated cloth shall conform to the quality and grade of product established by this specification. The occurrence of defects shall not exceed the applicable acceptable quality levels.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in contract, the contractor is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.1.1 Certificate of compliance. Where certificates of compliance are submitted, the Government reserves the right to check test such items to determine the validity of the certification.

4.2 Quality conformance inspection. Sampling for inspection shall be performed in accordance with MIL-STD-105, except where otherwise indicated hereinafter.

4.2.1 Components and material inspection. In accordance with 4.1, components and materials shall be inspected in accordance with all the requirements of referenced specifications, drawings, and standards unless otherwise excluded, amended, modified, or qualified in this specification or applicable purchase documents. In addition to the quality assurance provisions of applicable subsidary specification, the coating composition and coating distribution, shall be accepted on the basis of supplier's certificate of compliance with requirements specified for such characteristics.

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4.2.2 Examination of the end item. The end item shall be examined in accordance with the classification of defects at the inspection levels and acceptable quality levels (AQLs) set forth below. Random samples shall be drawn from each lot of end items of the same type, class and color for examination of visual and dimensional defects. The lot size for purposes of determining the sample size in accordance with MIL-STD-105 shall be expressed in units of yards for the examination in 4.2.2.1 and 4.2.2.2.

4.2.2.1 Yard-by-yard examination. The required yardage of each roll shall be inspected on one side only (however the side shall be alternated on every other roll examined), and visual defects classified as listed. The same yardage shall be given a through-light inspection for pinholes only as specified in MIL-STD-1487. The defects found shall be counted regardless of their proximity to each other except where two or more defects represent a single local condition of the cloth, in which case only the more serious defect shall be counted. A continuous defect shall be counted as one defect for each warpwise yard or fraction thereof in which it occurs. The sample unit shall be 1 linear yard. The acceptable quality level (AQL) for this examination shall be 1.5 major defects and 4.0 total defects per 100 units. The sample size shall be based upon inspection level III of MIL-STD-105. The number of rolls from which the sample is to be selected shall be in accordance with table II. An approximately equal number of yards shall be examined in each roll of the sample.

Defects	Classification Major	Minor
Hole, cut or tear, including edges--any.	X	
Pinholes--any--(except type IV cloth). 1/	X	
Abrasion or scratch--resulting in a thin or weak place or exposure of cloth structure.	X	
Blister, tunnel or delamination--any size.	X	
Lump, imbedded foreign matter or heavier coated area--exceeding 1/2 inch in combined length and width dimensions or area not uniformly coated. 1/		X
Evidence of dusting powder or talc.	X	
Crease or wrinkle--fold, pleat; doubling of surfaces that do not adhere to one another and that cannot be corrected by manual pressure.	X	

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Defects	Classification Major	Minor
Coating missing or noticeably thinner than normal.	X	
Coating noticeably heavier than normal.		X
Film missing, applied to lightly coated side, or thinner than normal. (Type I and II only)	X	
Window--due to absence or poor blending of pigmentation only.		X
Cloth edges rolled, folded or doubled.		X
Spot, stain or streak--one inch or more in combined directions. 1/	X	
--less than 1 inch, but greater than 1/2 inch in combined direction. 1/		X
Width (overall)--less than specified.		X
Matching (see 3.9.1), off shade, uneven, mottled.		X
Any tackiness or soft coating.		X
Laminate not uniform in thickness.		X
Fabric edges tight, resulting in clearly visible waviness or pucker on adjoining cloth.		X
Coating not heavier on face than on back.		X
Coating insufficient on back (bare threads exposed).		X
Any selvages not trimmed off when specified.		X
Cloth baggy: 2/ wavy or puckered--will not stay flat upon manual pressure.		X
Bowed filling 2/ more than two inches (measured from straight line cord to maximum point of arc).		X
Bias filling 3/ over three inches from horizontal at the greatest point of bias.		X
Objectional odor. 4/		X

1/ Clearly visible at normal inspection distance of 3 feet.

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2/ At the beginning of the examination, the cloth shall be laid out on a table at least 3 feet long by the full width of the cloth. One long side of the table shall be straight and one edge of the cloth shall be lined up with the straight side of the table. The cloth shall then be visually examined for the indicated defects.

3/ At the beginning of the examination, a cut shall be made 2 inches from the end of the roll and the cloth ripped across the roll to determine if the filling is bowed or bias.

4/ Aromatic odors usually associated with specific types of coating compounds or solvents shall not be regarded as objectionable.

TABLE II. Sample size (overall examination)

Lot size in yards	Sample size in rolls	Acceptance number
Up to 1,300 or less 1/	5	0
1,301 up to and including 3,200	7	0
3,201 up to and including 8,000	10	0
8,001 up to and including 22,000	15	0
22,001 up to and including 110,000	25	1
110,001 and over	35	1

1/ If lot contains fewer than 5 rolls, each roll in the lot shall be examined.

4.2.2.2 Examination for length.

4.2.2.2.1 Individual rolls. The sample unit shall be one roll. The number of rolls to be examined and the acceptance number shall be in accordance with table II. The following shall be considered a defect:

Any roll less than minimum or more than maximum specified.
Any roll containing more than 3 pieces.
Any piece in roll less than 6 yards.

4.2.2.2.2 Total yardage in sample. The lot shall be unacceptable if the total of the actual lengths of the rolls in the sample is less than the total of the lengths marked on the tickets.

4.2.2.2.3 Examination of preparation for delivery. An examination shall be made to determine that packaging and packing and marking requirements comply with section 5. The examination shall be conducted as specified in PPP-P-1136.

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4.2.3 Testing of the end item. The methods of testing specified in FED-STD-191, wherever applicable, and as listed in table IV shall be followed. The physical and chemical values specified in section 3 apply to the average of the determinations made on a sample unit for test purposes as specified in the applicable test method. The sample unit shall be 8 continuous yards, full width, of the finished cloth. The sample size shall be in accordance with table III. The lot will be unacceptable if one or more units fail to meet any requirement specified. For these characteristics where test results are reported as pass or fail each determination shall be reported and there shall be no evidence of failure of any test specimen to meet requirements as specified.

TABLE III. Sample size (end item)

Lot size (yards)	Sample size
800 or less	2
801 up to and including 22,000	3
22,001 and over	5

4.3 Test methods. Results of physical tests obtained under standard testing conditions defined in FED-STD-191, FED-STD-406 or FED-STD-601 will be acceptable except in case of dispute: in dispute cases, tests shall be conducted with both specimen and test apparatus under standard conditions as defined in FED-STD-191.

TABLE IV. Test methods

Characteristic	Requirement paragraph	Test method	Results reported as: Pass or fail	Results reported as: Numerically to nearest
Weight, oz./sq. yd.	3.6	5041 4/		0.1 ounce
Breaking strength	3.6	5102 4/		1.0 pound
Tearing strength:				
Initial	3.6	5132 4/		100 grams
After accelerated weathering	3.6	4.3.1.1 and 5132 4/		100 grams
After oven aging	3.6	4.3.1.2 and 5132 4/		100 grams
Hydrostatic resistance:				
Initial	3.6	4.3.2.1		1.0 psi
After strength of coating	3.6	4.3.2.2		1.0 psi

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TABLE IV. Test methods (cont'd)

Characteristic	Requirement paragraph	Test method	Results reported as: Pass or fail	Results reported as: Numerically to nearest
Film adhesion	3.6	4.3.3		0.1 pound
Coating adhesion	3.6	4.3.4		0.1 pound
Ply adhesion	3.6	4.3.5		0.1 pound
Adherability	3.6	4.3.6	X	
Blocking	3.6	5872 1/, 4/	X	
Distribution of coating	3.4	3/		
Polyethylene film (if used)	3.5	3/		
Resistance to chemical warfare agents Initial:				
Mustard - H	3.6	4.3.7		Minute
GB	3.6	4.3.7		Minute
After accelerated weathering:				
Mustard - H	3.6	4.3.7		Minute
GB	3.6	4.3.7		Minute
After decontamination:				
Mustard - H	3.6	4.3.7		Minute
GB	3.6	4.3.7		Minute
After cold crack	3.6	5874 2/, 4/ and 4.3.7		Minute
Air leakage:				
Standard condition	3.6	4.3.8	X	
At 130°F	3.6	4.3.8.1	X	

1/ Except that the test temperature shall be 200°F.

2/ The temperature shall be -40°F., and the film side of the cloth shall be on the outside of the fold. Specimen shall be visibly examined under 5 x magification for crack or delamination of the polyvinyl fluoride, and then forwarded to Aberdeen Proving Ground, Edgewood Area for H and GB resistance.

3/ A certificate of compliance shall be submitted and will be acceptable for the stated requirement.

4/ As defined in FED-STD-191.

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4.3.1 Tearing strength.

4.3.1.1 After accelerated weathering. Specimens shall be subjected to accelerated weathering test specified in Method 5804 of FED-STD-191 for a period of 200 hours without filters. The film side of the cloth shall be exposed. After exposure, the specimens shall be allowed to air dry and then brought to equilibrium under standard conditions prior to testing for tearing strength as specified in table IV.

NOTE: Attention is called to manufacturers who own weatherometers which operate at one cycle in two hours and with the water spray on continually. This equipment may continue to be used. However, the one cycle per minute shall be the standard procedure in the event of dispute, and the laboratory reports shall state the procedure used.

4.3.1.2 After oven aging. Specimens shall be exposed in a circulating air oven at 180°F. ± 2°F. for 100 hours, withdrawn form the oven and brought to equilibrium under standard conditions prior to testing for tearing resistance as specified in table IV.

4.3.2 Hydrostatic resistance.

4.3.2.1 Initial. The initial hydrostatic resistance test shall be conducted as specified in Method 5512 of FED-STD-191. The water pressure may be applied to either side of the laminated cloths. For the coated cloths the water pressure shall be applied to the lightly coated side of the cloth.

4.3.2.2 After strength of coating. Test specimens shall first be subjected to the strength of coating test in Method 5972 of FED-STD-191. These same specimens shall then be tested for hydrostatic resistance as specified in 4.3.2.1.

4.3.3 Film adhesion. The adhesion of the polyvinyl fluoride film to the chloroprene coating shall be determined by Method 5970 of FED-STD-191, except that the rate of jaw separation shall be 0.2 inches per minute. During the test, if the film is separated from the coating at an average force greater than that specified in table I or if the average force necessary to separate the cemented sample at the cement line is greater than that specified in table I, the adhesion of film requirement shall be considered as passing. If the film is separated from the coating, or the film and coating is separated from the cloth at an average force less than that specified in table I, the film adhesion requirement shall be considered as failing. If the average force necessary to separate the cemented sample at the cement line is less than that specified in table I, the test shall be considered as an improper bond and a solvent type adhesive shall be used. The original dry adhesion test shall be stopped after a three inch separation. The specimen shall be immersed in distilled water at room temperature for 24 hours, blotted dry, and the wet adhesion shall be determined on the remainder of the specimen.

4.3.3.1 Adhesion of film at elevated temperature. The adhesion shall be determined as outlined for the dry adhesion specified in paragraph 4.3.3, except that the area around the jaws of the testing machine and around the specimens shall be held at 180°F. ± 4°F. during the entire test period. The specimen shall be held at that temperature for not less than 15 minutes prior to start of the jaw separation.

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4.3.4 Coating adhesion. The adhesion of the chloroprene coating to the back side of the cloth shall be determined at standard conditions and at 180°F. as specified in Method 5970 of FED-STD-191.

4.3.5 Ply adhesion. The adhesion between the plies of the type II floor material shall be determined as specified in Method 5950 of FED-STD-191, except that the rate of jaw separation shall be 0.2 inches per minute. During the test, if the ply separates at the polyvinyl fluoride film-chloroprene coating line or the coating-base cloth line at an average force less than that specified in table I, the adhesion of the ply requirement shall be considered as failing. The original dry adhesion test shall be stopped after a 3-inch separation. The specimen shall be immersed in distilled water at room temperature for 24 hours, blotted dry and the wet adhesion shall be determined on the remainder of the specimen.

4.3.5.1 Ply adhesion at elevated temperatures. The adhesion shall be determined as outlined for the dry adhesion specimens in paragraph 4.3.5, except that the area around the jaws of the testing machine and around the specimens shall be held at 180°F. ± 4°F. during the entire test period. The specimens shall be held at that test temperature for not less than 15 minutes prior to the start of jaw separation.

4.3.6 Adherability test.

4.3.6.1 Preparation of sample. Two rectangular swatches approximately 8 inches x 10 inches of the laminated cloth shall be cut and joined by cementing to form a lap sample wherein the film side of one swatch is adhered to the chloroprene side of the other swatch. The warp directions of the swatches shall be parallel to their 10 inch dimensions. The lap seam shall be made by overlapping the 8 inch sides for a distance of 2 inches. The chloroprene coated area to be cemented shall be thoroughly abraded using 80 grit carborundum cloth, then washed with a toluol-wetted, clean cotton cloth. The film area to be cemented shall be throughly cleaned by wiping with toluol-wetted, clean cotton cloth. Allow cleaned surfaces to dry for 10 to 15 minutes at room temperature. To each surface, apply two coats of two part neoprene cement, Bostik 1095/Boscodur 5 (U.S.M. Chemical Company), (16/1 ratio by volume). Allow the first coat on each surface to dry 15 minutes. Five minutes after application of the second coat, assemble the seam and roll down by hand with a steel roller. Allow to cure at room temperature for seven days before testing as described below.

4.3.6.2 Specimens. One-inch wide strips shall be cut from the sample lap seam prepared as in 4.3.6.1, to provide 1 inch wide, approximately 18 inches long specimens for dead load tests described in 4.3.6.3. The strips shall be cut with their long dimensions perpendicular to the long dimension of the sample seam.

4.3.6.3 Dead load shear test.

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4.3.6.3.1 At standard conditions. A force of 40 pounds shall be applied lengthwise at one end of each of two specimens prepared as in 4.3.6.2 for four hours at standard conditions. For this test, one end of the specimen shall be anchored in a suitable grip. The anchoring grip and the force-application grip shall be of a type which precludes failures of the specimen at or in the grips, and they shall be applied not closer than 2 inches from the lapped seam. The grips shall hold the specimen uniformly across the one inch dimension, and the force shall be applied uniformly without twist. Suitable gage marks shall be made at the lap seam to assist in determining the amount, if any, of relative slippage of the lap seam at the conclusion of the test.

4.3.6.3.2 At 200°F. The test shall be conducted on a separate specimen in the manner as in 4.3.6.3.1, except in an oven at 200° ± 5°F.

4.3.7 Resistance to permeation by chemical warfare agents. All tests involving chemical warfare agents will be conducted by the Chemical Corps (see 6.4).

4.3.7.1 Resistance to mustard (H). Resistance to permeation by mustard H shall be determined as specified in Method 204 of MIL-STD-282. Four specimens shall be selected and four determinations made on the film side of the cloth. No test specimen shall be taken within 2 inches from the selvage edge. In the event that three of the four tested specimens meet the requirements, but one does not, then eight specimen samples shall be selected and one determination made on each specimen making a total of eight individual determinations. If seven of the eight specimens tested meet the requirements of table I and the eighth specimen tested is not less than 75 minutes, the material shall be considered acceptable.

4.3.7.2 Resistance to GB. Resistance to permeation by GB shall be determined as specified in Method 206 of MIL-STD-282. Test specimens shall be obtained by quartering a three yard piece of the coated cloth and selecting two samples, at random, from each quarter, with four determinations made on each side of the coated cloth. No test specimen shall be taken within two inches of the selvage edge. All samples shall meet the requirements listed in table I. In the event that seven of the tested specimens meet the requirements, but one does not, four additional test specimens from each quarter shall be tested (two on each side). If all sixteen specimens meet the requirements, the material shall be considered acceptable.

4.3.7.3 Resistance to mustard (H) after accelerated weathering. The test specified in 4.3.7.1 shall be repeated after exposing the specimens to be tested to an accelerated weathering for 48 hours as specified in Method 5804 of FED-STD-191 with the film side exposed.

4.3.7.4 Resistance to GB after accelerated weathering. The test specified in 4.3.7.2 shall be repeated after exposing the specimens to be tested to accelerated weathering for 48 hours as specified in Method 5804 of FED-STD-191 with the film side exposed.

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4.3.7.5 Resistance to mustard (H) after decontamination. The test specified in 4.3.7.1 shall be repeated after exposing the specimens to be tested to the decontamination procedure specified in 4.3.7.7.

4.3.7.6 Resistance to GB after decontamination. The test specified in 4.3.7.2 shall be repeated after exposing the specimens to be tested to the decontamination procedure specified in 4.3.7.7.

4.3.7.7 Decontamination procedure. Strips of laminated cloth, 10 by 22 inches, shall be cut along both the warp and filling directions. The strips shall be contaminated on the film side with liquid mustard (H) by brush application. A soft bristle paint brush 1-1/2 inches wide, shall be used. A total of 8 to 10 mg. of H shall be applied to each strip. Weight of H add-on shall be determined by weight loss measurements of the brush applicator. The strips shall then be suspended vertically, without wrinkling or folding, for one hour at room temperature. Each contaminated strip shall then be immersed for two hours in 4 gallons of boiling water while remaining in the suspended vertical position. After decontamination, the strips shall be dried in air at room temperature.

4.3.8 Air leakage at standard condition. Three specimens (13 inches in diameter) of the coated cloth shall be individually tested on the test jig as shown in figure I. A specimen shall be placed, film side up or lightly coated side down, on the holder and the plate collar tightly bolted thereon. Care should be taken to insure a leak-tight fit. The specimen shall then be inflated to an air pressure of 10 psi for 7 hours. After the 7 hour test period, the specimen shall be removed from the test jig and examined for the characteristics indicated in footnote 3 of table I.

4.3.8.1 Air leakage at 180°F. The test at elevated temperature shall be performed as specified in 4.3.8 except that infrared reflector type 250 watt, 115-125 volt heat lamp shall be so positioned that it will shine directly on the surface of the test specimen. The infrared lamp shall be turned on immediately when the specimen is inflated and height of the lamp adjusted above the specimen so the surface temperature of the specimen will not be less than 170°F, nor more than 220°F, in a 3 inch diameter area where the greatest concentration of infrared light contacts the specimen.

5. PACKAGING

5.1 Put-up and packaging. Put-up and packaging shall be level A or C as specified (see 6.2).

5.1.1 Levels A and C. The laminated cloth shall be put-up and packaged in accordance with the applicable requirements of PPP-P-1136.

5.2 Packing. Packing shall be level A, B and C as specified (see 6.2).

5.2.1 Levels A, B and C packing. The laminated cloth shall be packaged in accordance with the applicable requirements of PPP-P-1136.

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5.3 Marking. In addition to any special marking required by the contract or order, interior packages and shipping containers shall be marked in accordance with the applicable requirements of PPP-P-1136.

6. NOTES

6.1 Intended use. The coated cloth covered by this specification is intended for use in the manufacture of air supported shelter tents. The air supported shelter is defined as follows:

Type I, class 1 cloth is used for the outer layer and the olive green side of the cloth is exposed on the outside of the shelter.
Type I, class 2 cloth is used for the inner layer and the white coated side of cloth is exposed on the inside of the shelter.
Type II cloth is used for the floor of the shelter.
Type III cloth is used for the air-retaining web between the inner and outer walls of the shelter.
Type IV cloth is used for the non air-retaining web between the inner and outer walls of the shelter.

6.2 Ordering data. Procurement documents should specify the following:

(a) Title, number and date of this specification.
(b) Type and class of cloth required (see 1.2).
(c) Width of cloth required (see 3.7).
(d) Whether laboratory report approval is required (see 3.1).
(e) Selection of the applicable levels of packaging and packing (see 5.1 and 5.2).

6.3 Standard sample. For access to standard samples, address the procuring office issuing the invitation for bids.

6.4 Conductance of tests. Because of the extreme dangers involved in handling chemical warfare agents, and the special equipment required in connection with their use, all specification tests involving chemical warfare agents will be performed by the Edgewood Arsenal.

6.5 Adhesive - Bostik 1095. The adhesive shall not be more than 2 months old and shall show no stringiness of a cured or elastic nature. The cement must flow freely when brushed on any surface.

6.6 Curing agent - Boscodur 5. The curing agent shall not be more than 2 months old, and shall be stored in unopened cans or jars. The curing agent shall be measured by volume accurately, and immediately added to the already measured Bostik 1095. Care must be taken to keep the curing agent tightly covered except when being measured.

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6.7 Supersession data. This specification supersedes the requirements of Limited Production Purchase Description LP/P. DES 11-71, dated 30 June 1971, Cloth, Laminated, and Cloth, Coated (Chloroprene Coated).

Custodians:

Army - GL
Navy - SA
Air Force - 11

Review activities:

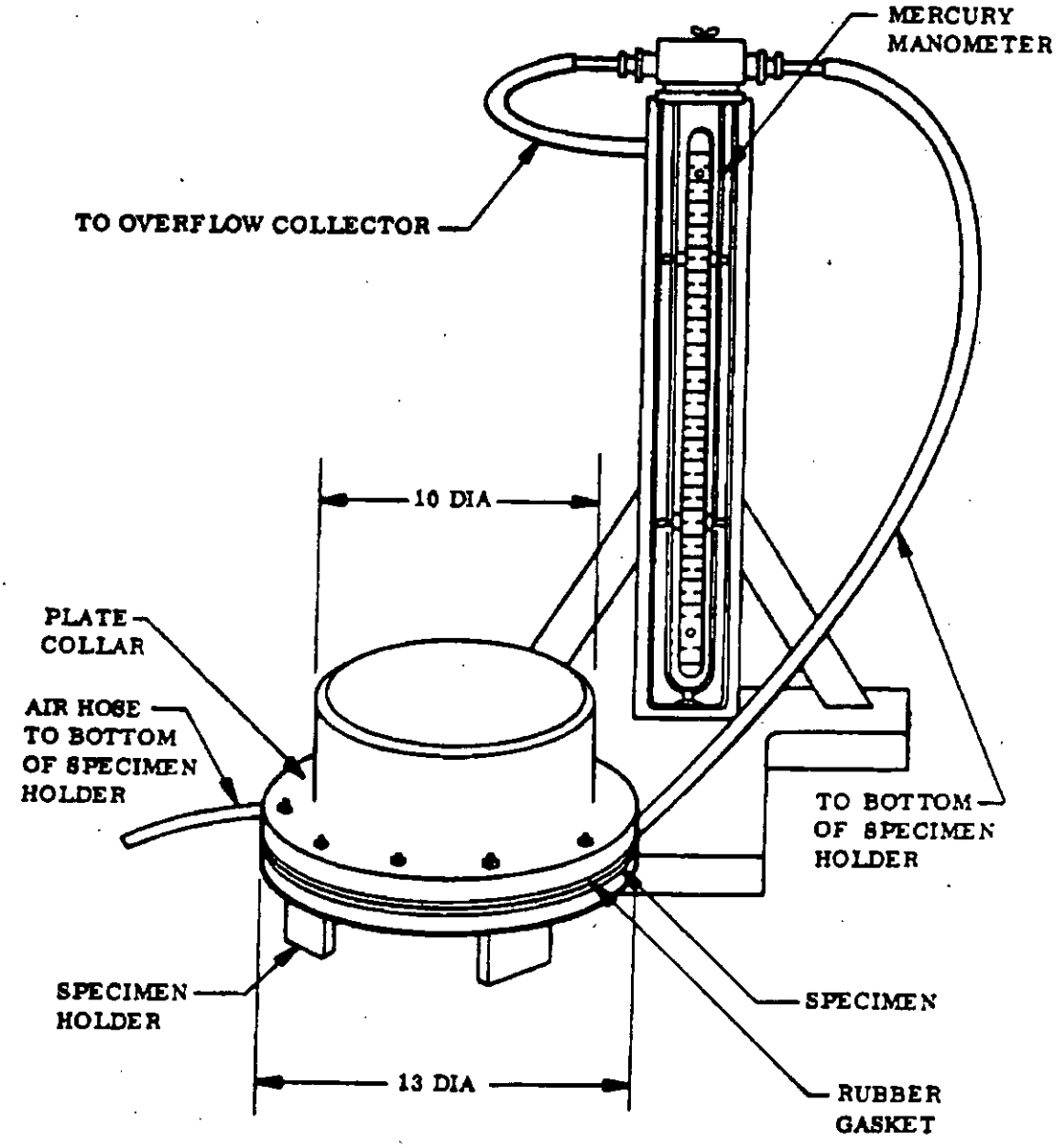
Army - EA, MD
Air Force - 82
DSA - CT

Preparing activity:

Army - GL

Project No. 8305-0502

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DIMENSIONS IN INCHES

FIGURE 1. AIR LEAKAGE TEST JIG

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STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

OMB Approval
No. 22-R255

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DOCUMENT IDENTIFIER AND TITLE MIL-C-43944 CLOTH, LAMINATED, AND CLOTH (CHLOROPRENE COATED)

NAME OF ORGANIZATION AND ADDRESS

CONTRACT NUMBER

MATERIAL PROCURED UNDER A

☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

1. HAS ANY PART OF THE DOCUMENT CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?

A. GIVE PARAGRAPH NUMBER AND WORDING

B. RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES

2. COMMENTS ON ANY DOCUMENT REQUIREMENT CONSIDERED TOO RIGID

3. IS THE DOCUMENT RESTRICTIVE?

☐ YES ☐ NO *(If "Yes", in what way?)*

4. REMARKS

SUBMITTED BY *(Printed or typed name and address - Optional)*

TELEPHONE NO

DATE

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